

PURPOSE & PATTERNS

Model infectious disease spread (SIR) to evaluate NPI impact and support Digital Twin decision-making.

DATA INPUTS

Census Datasets

STATIC

Population size, age, gender, occupation
-> Agent initialization (N, demographics)

GPS Mobility Data

DYNAMIC

Location traces; work/school/home context
-> Agent mobility patterns & contact rates

Public Health Reports

DYNAMIC

Confirmed cases, β , γ estimates
-> Infection initialization; parameter calibration

DATA PIPELINE

1. Collection

Local gov. census API | GPS provider feed | Health dept. reports

2. Pre-processing

Multiple imputation (missing counts) | Outlier removal | QC checks

3. Analysis

Statistical estimation of β & γ | Contact-rate derivation from GPS

ABM CORE: SIR Epidemic ABM

AGENTS (n=[N = population])

Types: Individual

State variables:

- disease_state (S/I/R)
- age
- gender
- occupation
- location
- adherence_prob

-> Census + Public Health Reports

Submodels (Behaviors)

Movement

Agent moves across home/work/school

Transmission

$S \rightarrow I$ on contact with infected agent

Recovery

$I \rightarrow R$ at rate γ ; permanent immunity

Compliance

Stochastic NPI adherence from survey dist.

ENVIRONMENT

Grid: Spatial grid

Resolution:

Location-level
Layers: Home, Work, School

-> GPS Mobility Data (location layers)

INTERACTIONS

* Agent <> Agent

Disease transmission: $S \rightarrow I$ on contact (prob = $\beta \times$ adherence)

* Agent <> Env

Mobility: agents move between home / work / school locations
NPI encoding: β multipliers applied per location context

* Topology

Spatial grid with location layers
Contact within co-located agents each time step

-> GPS Mobility Data; Public Health Reports

Temporal: 1 day | Duration: Until epidemic ends

Stop: $I = 0$ OR max time steps reached

OBSERVATIONS

[~] Epidemic Curve

$S(t)$, $I(t)$, $R(t)$ counts over time
Time series
Emergent: Peak timing & magnitude from agent interactions

[=] NPI Impact

Δ peak infections across intervention scenarios
Comparative bar / curve
Emergent: Emergent herd-immunity threshold

MODEL EVALUATION

Calibration

Method: Bayesian inference (MCMC)
Params: β , γ
Data: Historical infection time-series
Result: Calibrated β^* , γ^* with 95% CI

Validation

- Output validation: RMSE/MAE vs held-out surveillance data
- Scenario validation: simulated vs post-NPI surveillance trends
- Face validity: expert review of epidemic curve shape

Result: RMSE < 5% of peak; expert-approved

SCENARIOS (if applicable):

Baseline

No NPI; calibrated β^* , γ^*

Social Distancing

$\beta \times (1 - \text{efficacy})$

Vaccination

$S \rightarrow R$ at rollout rate

★ Combined NPI

Distancing + masking + vaccination; key DT policy scenario